Expression Evaluator with Scientific Calculator

Abstract

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The Expression Evaluator with Scientific Calculator is our Data Structures project developed using the C programming language. The main objective of this project is to evaluate mathematical expressions containing arithmetic operators, variables, and scientific functions. The project uses stacks and arrays as its primary data structures to perform expression conversion and evaluation efficiently.

The program converts an infix mathematical expression into postfix notation using the stack data structure. During conversion, operator precedence and parentheses are handled to ensure the correct order of operations. The postfix expression is then evaluated using a separate stack that stores numerical values. The program supports arithmetic operations such as addition, subtraction, multiplication, division, modulus, and exponentiation. It also supports scientific functions, including sine, cosine, tangent, square root, and logarithm. Trigonometric functions accept angles in degrees, and the program allows users to enter values for variables during evaluation.

Arrays are used to store the input expression, operator stacks, postfix expression, and numerical results, while stack operations such as push and pop are used during expression conversion and evaluation. The project demonstrates important data structures concepts, including stack implementation using arrays, infix-to-postfix conversion, operator precedence, and postfix expression evaluation. Overall, this project provides a practical application of data structures in mathematical computation and demonstrates how stacks can simplify the processing and evaluation of complex expressions.